Introduction: from belief sharing to robust generalized Bayes

Introduction: from belief sharing to robust generalized Bayes I

A colony of active-inference agents that shares beliefs inherits both the power and the fragility of the pool it uses: the same multiplication of reports that sharpens an honest consensus lets a single confident, wrong member capture it. Two research communities each hold part of the remedy but do not, on their own, close the gap. The active-inference community gives agents generative models, action selection, and colony-level belief sharing. The robust- and federated-Bayes community gives generalized objectives and aggregation methods for inference under misspecification or contamination. The cited literatures do not, however, provide a single tested treatment of robust categorical belief fusion for active-inference agents. This introduction states the problem and its evidence boundary; sec. 3 scopes the reviewed gap, and sec. 4 lists the contributions together with what each result does not establish.

Active inference supplies generative agents and shared beliefs

Active inference supplies generative agents and shared beliefs I

Active inference casts perception, learning, and action as the minimization of variational free energy under a generative model, a program that grew out of the free-energy principle (Friston 2010) and its process-theory implementation (Friston et al. 2017). In the discrete state-space setting the formalism is now standardized: an agent carries the categorical tensors A (likelihood), B (transitions), C (preferences), and D_0 (initial priors), infers hidden states by minimizing variational free energy, and selects actions by minimizing *expected* free energy — the sum of a risk term and an ambiguity term that together trade off goal-seeking against uncertainty-resolving behavior (Da Costa et al. 2020). This synthesis is the substrate we build on. Mature toolboxes make it executable at scale: pymdp provides discrete-state active inference in Python (Heins et al. 2022), and

Active inference supplies generative agents and shared beliefs I

RxInfer delivers reactive message passing for exact Bayesian inference (Bagaev et al. 2023).

The community has also moved from single agents to *collectives*. Surprise-minimizing ensembles reproduce collective animal behavior such as schooling and flocking (Heins et al. 2024); epistemic communities form when agents share a generative model and exchange evidence (Albarracin et al. 2022); and collective intelligence has been framed directly in active-inference terms (Kaufmann et al. 2021). The narrow bridge studied here starts from posterior broadcasts over one shared categorical factor — a predator's location, say — and forms a project log-linear pool (eq. 6). Under the explicit finite-shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 5.8, that weighted geometric pool is a specialization of Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024). It is not a reconstruction of the source message construction, scheduling, cavity policy, generative factors, or complete protocol. That representation connects the qualified categorical bridge to the classical logarithmic-pooling literature (Genest and Zidek 1986; Genest et al. 1986), to modern Bayesian treatments of log-pool weights (Carvalho et al. 2023), to product-of-experts geometry in machine learning (Hinton 2002), and to distributed Bayesian estimators (Tresp 2000). In the reproduced baseline, communication lowers mean free energy relative to the matched incommunicado condition (sec. 7.2). Friston et al. (2024) crystallized the colony mechanism into three worked simulations — communicating-colony free-energy convergence, Dirichlet language acquisition, and Bayesian model reduction structure emergence — whose mechanisms motivate three reduced categorical analogues in this repository (sec. 7.1). They are not numerical or exact protocol replications; a source-parity reconstruction remains future work before evaluating source-level equivalence. Alongside fusion, the community has tools for growing the model itself: active inference connects naturally to Bayesian optimal experimental design and model selection (Smith et al. 2022), and post-hoc Bayesian model reduction prunes redundant structure by comparing free-energy bounds (Friston and Penny 2011) — the engine behind our emergence study (sec. 7.4).

Active inference supplies generative agents and shared beliefs I

The reviewed active-inference belief-sharing work does not systematically characterize fusion under explicit contamination or intentionally wrong belief broadcasts. The log-linear pool assumes reports are compatible with the shared generative model, while the opinion-pooling literature makes the assumptions about independence, weights, and external Bayesian coherence explicit (Genest and Zidek 1986; Genest et al. 1986). Fully Bayesian aggregation sharpens the point: geometric pooling is normatively compelling under dynamic-Bayesian rationality assumptions, not an assumption-free robustness procedure (Dietrich 2021). The same product geometry can be brittle: if a report assigns zero or near-zero mass to the true state, the product can assign near-zero mass there too. That is the failure mode tested here, not a claim that every federation or every product pool behaves identically.

Robust and federated Bayes supplies bounded-influence updating

A separate literature studies bounded-influence updating outside active inference. **Federated learning** aggregates models trained on decentralized data without pooling the data itself, the canonical algorithm being FedAvg (McMahan et al. 2017). Cast probabilistically, federated and continual learning unify under **partitioned variational inference**, in which each client owns a factor of a global approximate posterior and the server combines factors in natural-parameter space (Bui et al. 2018; Ashman et al. 2022). This is a closely related factor algebra expressed in a different vocabulary; the implementation tests the correspondence in the categorical recovery limit rather than assuming that the two settings are interchangeable.

Robust and federated Bayes supplies bounded-influence updating I

The robustness this paper needs comes from **generalized Bayesian inference**, which replaces the likelihood with a *loss* and the KL regularizer with a *general divergence*, so that the posterior minimizes a generalized objective rather than applying Bayes' rule literally (Bissiri et al. 2016; Jewson et al. 2018; Knoblauch et al. 2022). This includes Gibbs-posterior updates (Jiang and Tanner 2008), coarsened or tempered posteriors that condition on neighborhoods rather than exact data (Miller and Dunson 2018), and learning-rate/temperature choices designed for safe updating under misspecification (Grünwald 2012; Kleijn and Vaart 2012). Choosing a loss or divergence with a bounded-influence property — for example, the density-power β-loss (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015) or generalized cross-entropy (Zhang and Sabuncu 2018) — can cap the influence of a contaminated observation under the corresponding assumptions. **FedGVI** (Mildner et al. 2025) federates this idea: each client runs a robust generalized-Bayes update against a *cavity* (the global posterior with the client's own factor removed), and the server aggregates the refreshed factors under a chosen server divergence. The result is federated inference with divergence- and loss-specific robustness guarantees, demonstrated on Bayesian neural networks under label contamination.

Crucially, standard Bayes is a *corner* of this generalized family — the case where the loss is the negative log-likelihood and the divergence is KL. Friston et al. do not claim FedGVI, generalized Bayes, β-divergence, or robustness; this manuscript supplies the recasting. Robust generalized Bayes therefore does not replace exact-Bayes belief fusion; it contains the standard pool as a tested zero-robustness recovery limit. That containment is the hinge this paper tests, and the recovery limits are stated formally as numbered results in sec. 6 and the central identity in sec. 5.8.

The historical framing is deliberately modest. The manuscript uses early probability, inverse-probability, utility, and collective-judgment sources as a conceptual genealogy for belief, evidence, expectation, and aggregation (Pascal and Fermat 1654; Huygens 1657; Bernoulli 1713; Moivre 1718; Bayes 1763; Laplace 1774; Bernoulli 1738; Condorcet 1785). These sources do not anticipate KL divergence, product-of-experts learning, variational Bayes, or federated optimization; the modern correspondence to FedGVI is the formal construction proved and tested here.

Beyond this core, we evaluate a tempered objective family, a deterministic MLP aggregation transfer, and disjoint-field-of-view communication. These are boundary tests: the first probes the accuracy/weight-control trade-off, the second tests API portability in one additional model class, and the third separates a binary-complement null result from a larger-state-space case where communication materially improves consensus.

Questions, design, and evidence boundary

Questions, design, and evidence boundary I

The paper answers four scoped questions. First, does turning robustness off recover the standard log-linear pool and closed-form Bayes update? Second, under the declared confident-wrong broadcast mechanism, how does the server heuristic change consensus accuracy across contamination rates? Third, what does the objective-backed variational server rule guarantee, and what accuracy does it trade away? Fourth, how do communication, hierarchy, sensitivity, and parameter recovery behave in the accompanying categorical extensions? The first question is answered algebraically and by machine-precision checks; the others are conditional simulation results. The independent unit, resampling scheme, and fixed hidden-state/attack-target estimand are specified in sec. 5.22 and sec. 5.27.

How to read the visual architecture I

The manuscript uses two complementary visual layers. The formal schematics adapt the generative-model and posterior-sharing perspective of Friston et al. (2024) to the categorical implementation: fig. 4 shows the private sensory report and $A/B/C/D$ substrate, fig. 3 shows how three local posterior messages become server inputs, and fig. 5 shows the hidden-state, agent, and active-control context. These diagrams are explanatory maps, not additional empirical observations. The data-bearing figures then report the executed recovery checks, conditional contamination sweep, and objective/descent diagnostics; their captions identify the relevant uncertainty and resampling unit. The graphical abstract in fig. 1 compresses the same logic into a recovery anchor, a federation pathway, and three explicitly non-transferable robustness axes.

fig. 1 illustrates one configured failure-and-repair case: a partially contaminated colony broadcasts beliefs (Panel A), the equal-weight log-linear pool is pulled toward the attack target (Panel B), and the server-side heuristic reweights the broadcasts (Panel C). It is a deterministic schematic, not a universal robustness claim; the three axes and their distinct guarantees are defined in sec. 3 and sec. 3.3.

How to read the visual architecture I

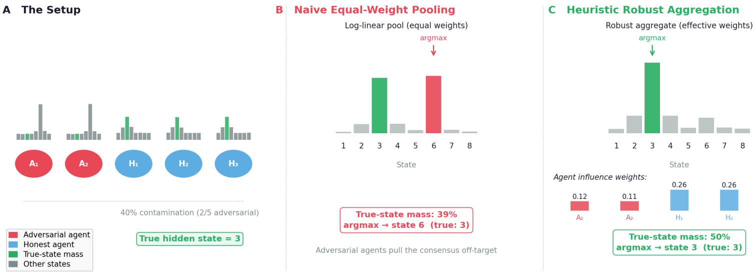

Figure 1: System overview. Source relation: original project schematic; estimand: displayed posterior-mass and influence-weight contrasts; uncertainty: none. 5 agents with heterogeneous beliefs (blue = honest, red = adversarial) feed into naive pooling (Panel B, argmax pulled off-target) versus canonical `robust_aggregate` heuristic reweighting (Panel C, true state recovered). x-axis is hidden-state index (1–8) in Panels B and C; y-axis: bar height indexes probability mass per state, and each agent in Panel A carries its own mini posterior bar chart (green column = true state). The true hidden state is 3; under 40% contamination the equal-weight pool concentrates 39% of consensus mass on the true state (its argmax lands on the adversaries' state), while the heuristic concentrates 50% and recovers the correct argmax. This is a single deterministic schematic (no resampling, hence no error bars or CI); all percentages are computed from the pooled beliefs shown, and the panel does not claim the variational server's objective-backed weight-control result for `robust_aggregate`.